

Journal 06: Security and Transparency Risks in AI Systems

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Artificial intelligence (AI) continues to permeate all aspects of life and is especially prevalent in software engineering and code generation. With the rise of powerful AI code assistants such as Github Copilot, Claude Code, Tabnine, and others, vigilance surrounding security, privacy, transparency, and bias is vital. This paper first explains the security risks of AI-generated code and the role of transparency, before exploring the vulnerabilities and ethical challenges associated with these risks. Ideas for risk mitigation strategies are provided, and the responsibilities of different entities in establishing and enforcing AI security and transparency standards are discussed. Ultimately, addressing these challenges requires a combination of transparent AI systems, responsible development practices, and coordinated oversight from users, developers, and governments.

AI code and productivity tools introduce many security and ethical risks. Models often regurgitate data from training sets that include unsecure coding practices, hardcoded secrets or credentials, and poor encryption (Sheggam & Zhang, 2024). This presents a major concern for the security of AI-generated code. To exacerbate this issue, models often hallucinate and provide incomplete or inaccurate information, sometimes up to 50% of the time (Liu et al., 2025). When AI is used for decision making it is vital to understand how the decision was reached. In other words, the model should be transparent. Without this openness, it becomes difficult or impossible to properly assess the output and determine its validity. Transparency is also important for ensuring proper data integrity and privacy. Transparent models should be open about what data they use and how they use it, allowing users to opt into data collection and decide whether the downstream use of their data is appropriate (Herbold et al., 2024; Lucchi, 2023). This increases public trust and confidence, leading to greater adoption and acceptance of AI tools.

In addition to security risks, AI code tools present significant ethical challenges. Privacy must be considered for both the input and training data used for models, and intellectual property rights also affect model output (Lucchi, 2023). Proper adherence to software licensing remains an open challenge, specifically for copyleft licenses. Current models are incapable of providing accurate licensing information about the code they generate, despite being near verbatim copies (Xu et al., 2025). Additionally, AI tools can introduce compounding issues in the form of ethical debt, a concept modeled off technical debt (Salah, 2025). Ethical debt occurs when “normative concerns are deferred or overlooked in favor of speed, functionality, or innovation” (Salah, 2025). For example, an engineer may accept seemingly benign code suggestions without assessing the underlying privacy or security risks. Alternatively, outdated and insecure coding practices may remain in use because time constraints do not allow for proper review. To compound this issue further, responsibility erodes and authorship becomes diluted, as many programmers view AI tools as authoritative coauthors (Salah, 2025).

To combat security risks, both users and model developers should implement mitigation strategies. From the user’s perspective, proper management, review, and oversight can reduce the amount of accrued ethical debt and help catch potential security issues (Salah, 2025). Toolchain integrations that automatically review code through static analysis for security, liability, and privacy violations can reduce human workload and make these practices more feasible for resource-constrained groups (Salah, 2025). Users should also consider maintaining local versions of their code and data—stored with proper encryption—that are separate from versions used with third-party tools and processing (Sheggam & Zhang, 2024). Credentials, personally identifiable information, trade secrets, and any other sensitive data should be scrubbed before being exposed to the AI assistant (Sheggam & Zhang, 2024). Model developers must also implement mitigation

strategies, including safeguards against algorithmic bias in training data, manipulation intended to expose protected data, and methods to ensure their tools are not used for harmful actions such as deepfakes, political propaganda, or financial scams (Liu et al., 2025).

However, implementing these mitigation strategies requires coordinated efforts from multiple stakeholders; the responsibility for enforcing security and transparency standards does not fall on a single group. Rather, users, developers, and governments must all contribute to the safe usage of AI. As discussed, users should implement proper review and management to catch latent security and privacy violations. Developers must incorporate algorithms, filtering, or other mitigation techniques to prevent algorithmic bias and harmful use. Governments are responsible for publishing guidelines for both users and developers and for navigating the balance between restriction of innovation and data privacy and integrity. For example, the United States has published the Blueprint for an AI Bill of Rights, which includes five basic principles such as bias protection and right to notice and explanation (Liu et al., 2025).

These shared responsibilities highlight the broader societal implications of AI-assisted software development. In conclusion, security and transparency risks of AI tools impact everyone from individual users to governments. Proper mitigations and safeguards must be implemented to protect against cyberattacks and malicious usage. Proper review, management, and oversight are needed to reduce the accumulation of ethical debt and ensure both data privacy and intellectual property rights. Transparent models are vital for ensuring that decisions are understandable and explainable, leading to greater public trust and accountability in the use of AI systems.

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